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Roll No

IT-5001-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

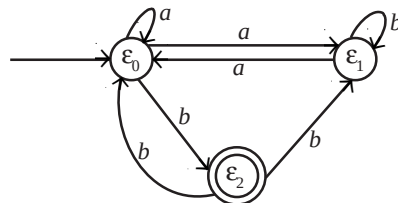
Theory of Computation

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain finite automation and its various types.
b) Design DFA that accepts all strings with at most 3 a's?
2. a) Construct a DFA equivalent to NFA



- b) Distinguish between Moore and Melay Machines.
3. a) State and prove the pumping lemma theory of Regular language.
b) Define Two way finite Automata with suitable example.
4. a) Define left most derivation, right most derivation and parse tree with suitable example.
b) Convert the following grammar into GNF.
 $S \rightarrow AA / 0$
 $A \rightarrow SS / 1.$

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5. a) Let G be the grammar
 $S \rightarrow OB|IA, A \rightarrow O|OS|IAA, B \rightarrow I|IS|OBB$
For the string 00110101 Find:
i) LMD
ii) RMD
iii) Parse tree
b) If G is the Grammar $S \rightarrow SbS|a$ show that G is ambiguous.
6. a) Design a TM to perform 2's complement operation on binary string. Show the IDs for the string 010 and 101.
b) Construct a TM for the language of even number of 1's and even number of 0's over $\Sigma = \{0,1\}$.
7. a) What do you mean by Derivation tree? Give examples.
b) What do you mean by Universal Turing Machine (UTM)?
What are applications of it?
8. Write short notes (any three):
a) Recursive and Recursively enumerable language
b) Halting problem
c) NP complete Vs NP hard
d) CNF
e) NPDA
